

Prelim I
Survey Open.
Take Today!

Properties of Rigid Objects

- Extensive - Mass
 - Linear Momentum
 - Angular Momentum
- Angular Velocity

HW4 due
Today 6pm
(Late until 12)

Rigid Objects

- We can't do simulations of 10^{23} particles
- Closed systems, where mass doesn't change.
- Overall motion $\gg$ deformation.

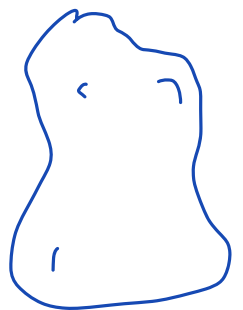
Extensive Properties

$$m = \sum m_i \quad \text{or} \quad \int dm$$

$$\vec{p} = \sum \vec{p}_i \quad \text{or} \quad \int \vec{v} dm$$

$$\vec{H}_{/I} = \sum \vec{H}_{i/I} \quad \text{or} \quad \int \vec{r}_{dm/I} \times \vec{v}_{/I} \cdot dm$$

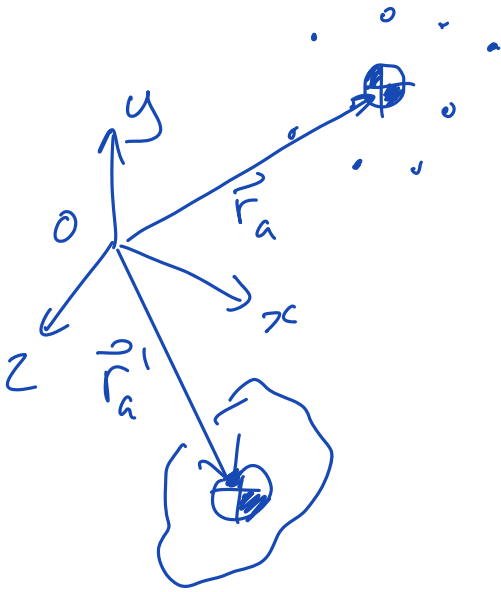
$$E_k = \sum E_{k_i} \quad \text{or} \quad \frac{1}{2} \int \vec{v}^2 dm$$



Mass

The Center of Mass (CoM) $\Rightarrow$ $\odot$ or $\oplus$

$\Rightarrow$ Average position of mass $\vec{r}_a = \frac{\vec{r}_1 m_1 + \vec{r}_2 m_2 + \dots + \vec{r}_n m_n}{m_{\text{tot}}}$



$$m_{\text{tot}} \vec{r}_a = \begin{cases} \sum m_i \vec{r}_i \\ \int \vec{r}_{dm} dm \end{cases}$$

$$m \vec{r}_a = \int \vec{r} dm$$

Linear Momentum

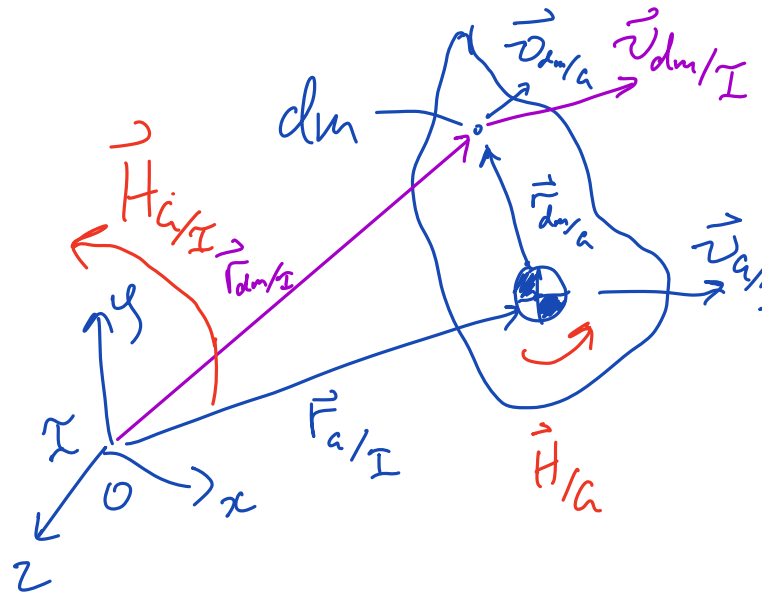
$$\vec{p} = \begin{cases} \sum m_i \vec{v}_i \\ \int \vec{v} dm \end{cases} = \int \frac{d\vec{r}}{dt} \cdot dm = \frac{d}{dt} \int \vec{r} \cdot dm$$

$$\vec{p} = m \frac{d\vec{r}_a}{dt} = m \vec{v}_a$$

$$\Rightarrow \vec{\dot{p}} = m \vec{a}_a$$

Angular Momentum

$$\vec{H}_I = \begin{cases} \sum \vec{r}_{i/I} \times m_i \vec{v}_{i/I} \\ \int \vec{r}_{dm/I} \times \vec{v}_{dm/I} \cdot dm \end{cases}$$



$$(\vec{r}_{a/I} + \vec{r}_{dm/a}) \times (\vec{v}_{a/I} + \vec{v}_{dm/a})$$

Expand components of cross product.

$$\vec{H}_I = \int \vec{r}_{a/I} \times \vec{v}_{a/I} \cdot dm + \int \vec{r}_{dm/a} \times \vec{v}_{a/I} \cdot dm + \int \vec{r}_{a/I} \times \vec{v}_{dm/a} \cdot dm + \int \vec{r}_{dm/a} \times \vec{v}_{dm/a} \cdot dm$$

$$= \vec{H}_{a/I} + \int \vec{r}_{dm/a} \cdot dm \times \vec{v}_{a/I} + \vec{r}_{a/I} \times \int \vec{v}_{dm/a} \cdot dm + \vec{H}_{a/a}$$

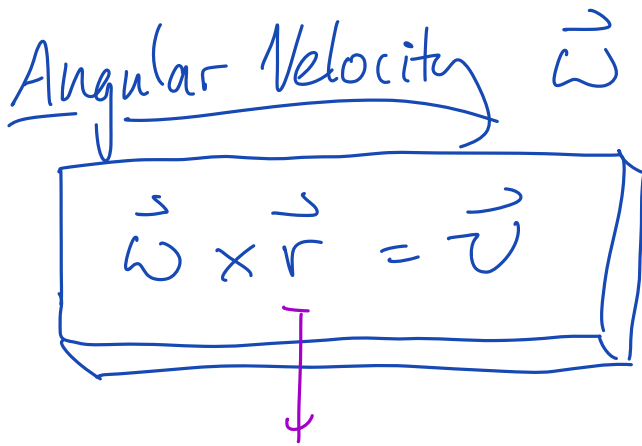
$$\int \vec{r}_{dm} - \vec{r}_a \cdot dm = 0$$

$$\int \vec{v}_{dm} - \vec{v}_a \cdot dm = 0$$

i.e. The weighted average of all points/velocities w.r.t to CoM must be zero.

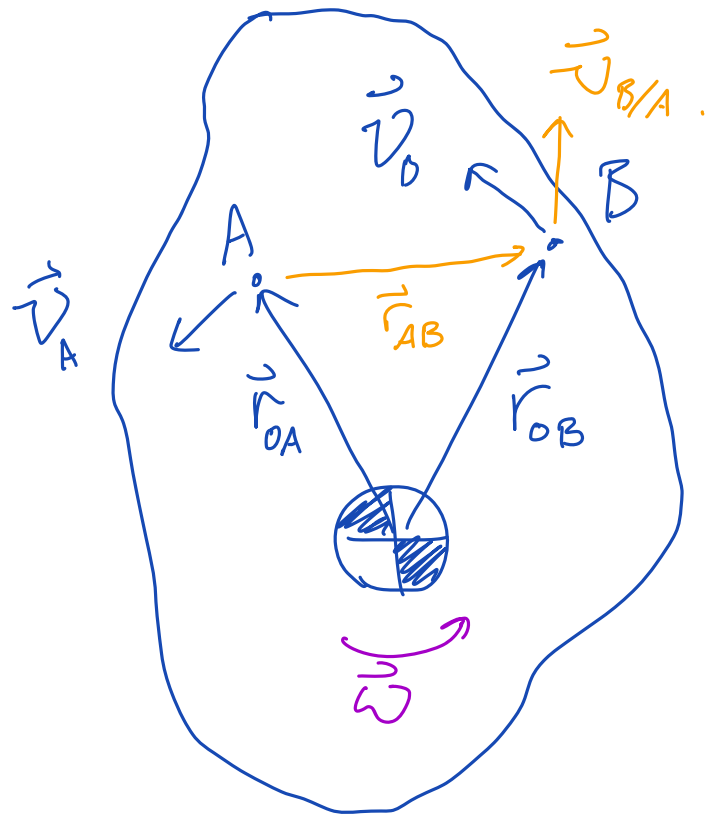
$$\boxed{\vec{H}_I = \vec{H}_{a/I} + \vec{H}_{a/a}}$$

$\Rightarrow$ See HW5 for E_k .



any line segment
within the rigid object.

Consider $\vec{H}_A$



$$\vec{H}_A = \int \vec{r}_{dm/A} \times \vec{v}_{dm/A} dm$$

$$= \int \vec{r}_{dm/A} \times (\vec{\omega} \times \vec{r}_{dm/A}) dm$$

In 2D

$$= \omega I^A \hat{k}$$



Polar moment of Inertia

$$I^A = \int \vec{r}_{dm/A}^2 dm$$

